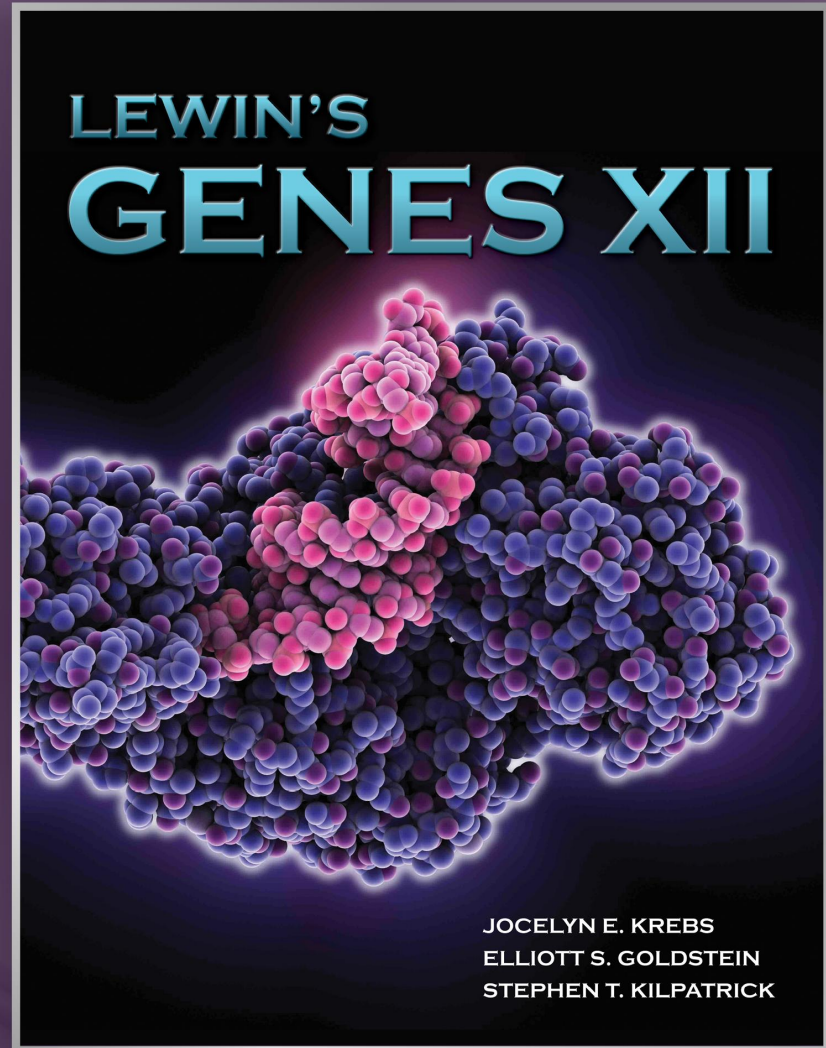
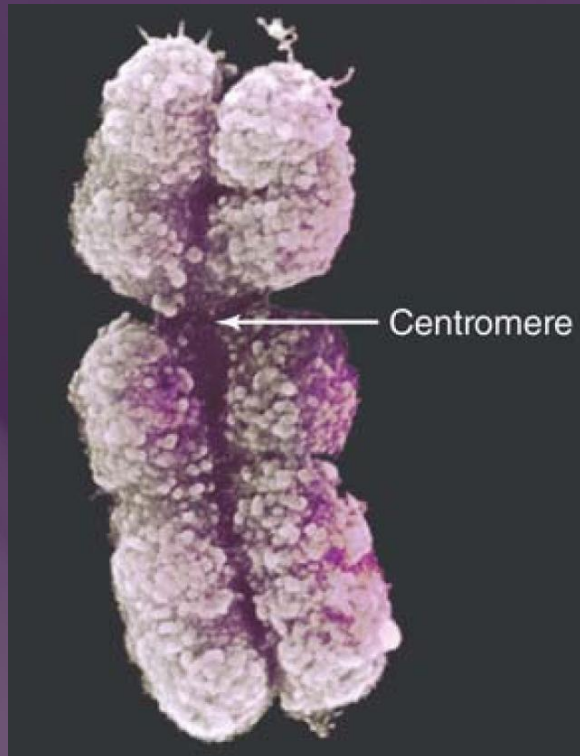


Chapter 3

Chromosomes & Chromatin



3.1 Introduction

- **chromosome** – A discrete unit of the genome carrying many genes. Each consists of a very long molecule of duplex DNA and an approximately equal mass of proteins.
 - It is visible as a morphological entity only during cell division.
- **nucleoid** – The structure in a prokaryotic cell that contains the genome.
 - The DNA is bound to proteins and is not enclosed by a membrane.

3.2 Viral Genomes Are Packaged into Their Coats

- **capsid** – The external protein coat of a virus particle.
- Filamentous RNA viruses condense the RNA genome as they assemble the head shell around it.
- Nucleic acid within the head shell is extremely condensed.
- **nucleation center** – A duplex hairpin in TMV (tobacco mosaic virus) in which assembly of coat protein with RNA is initiated

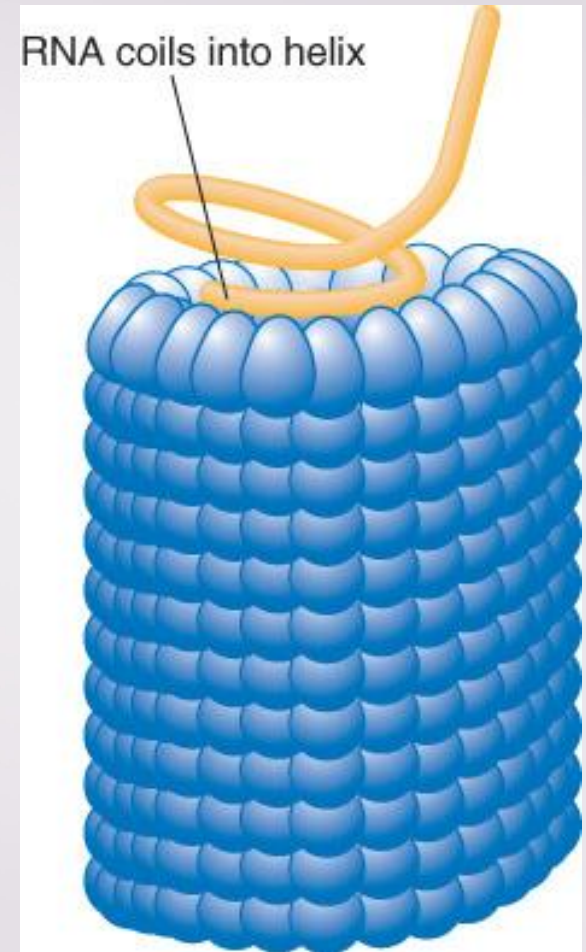


Figure : A helical path for TMV RNA is created by the stacking of protein subunits in the virion (the entire virus particle).

3.2 Viral Genomes Are Packaged into Their Coats

Spherical DNA
viruses insert the
DNA into a
preassembled
protein shell.

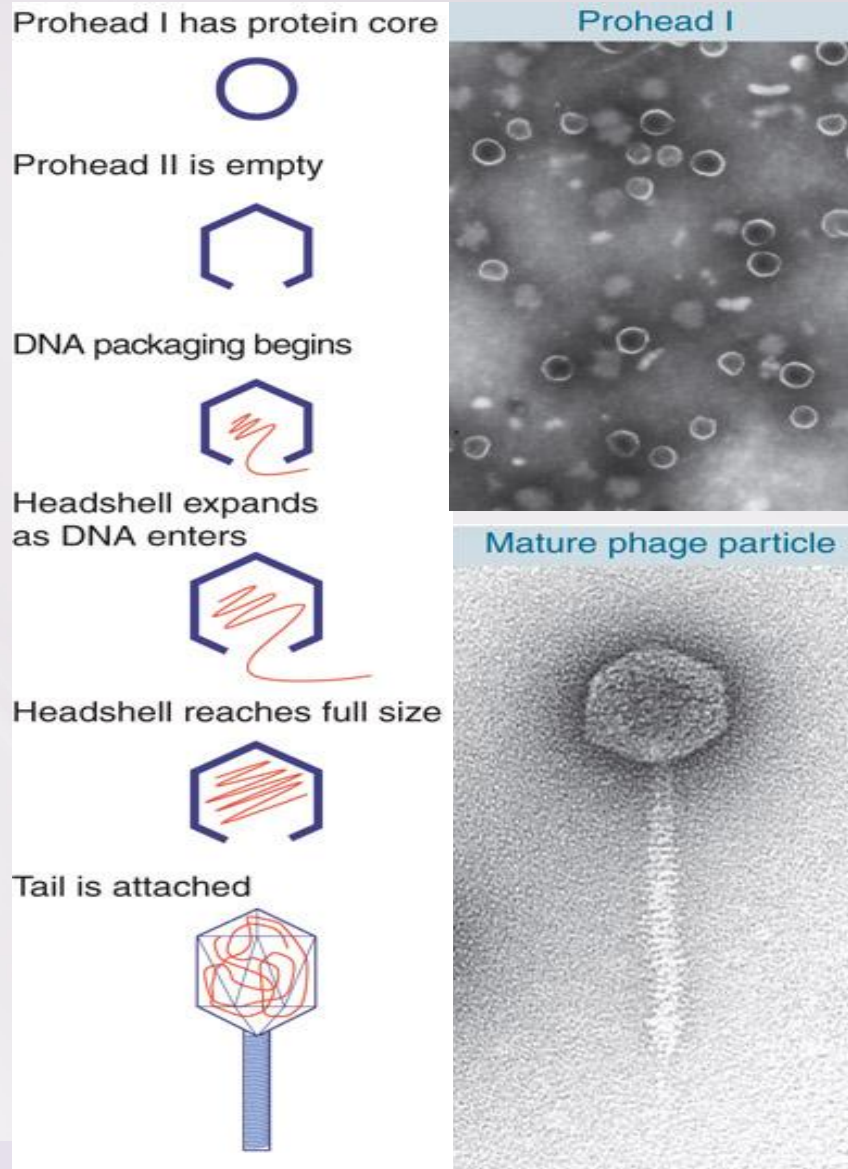


Figure : Maturation of phage lambda passes through several stages.

3.2 Viral Genomes Are Packaged into Their Coats

- **terminase** – An enzyme that cleaves multimers of a viral genome and then uses hydrolysis of ATP to provide the energy to translocate the DNA into an empty viral capsid starting with the cleaved end.

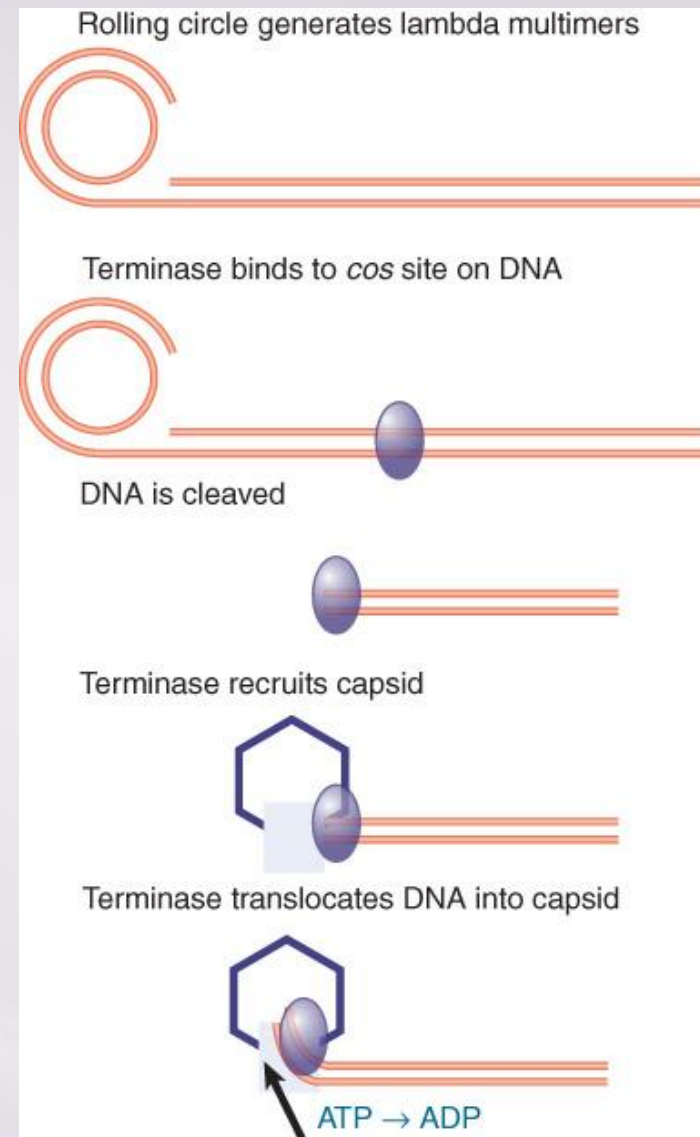


Figure : Terminase protein binds to specific sites on a multimer of virus genomes generated by rolling circle replication.

3.3 The Bacterial Genome Is a Nucleoid with Dynamic Structural Properties

- The bacterial nucleoid is organized as multiple loops compacted by nucleoid-associated proteins such as H-NS and HU.
- **Nucleoid-associated proteins** are typically small, abundant DNA-binding proteins that function in nucleoid architecture, domain topology, and gene regulation.

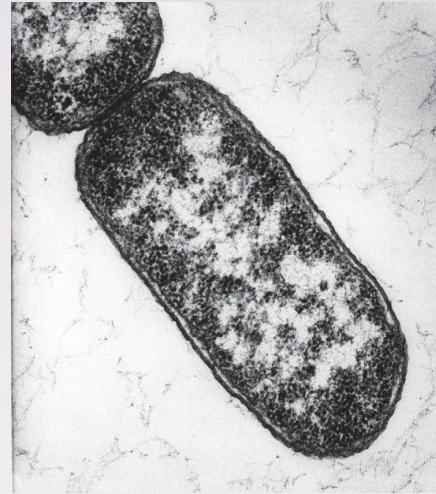


Figure A: A thin section shows the bacterial nucleoid as a compact mass in the center of the cell.



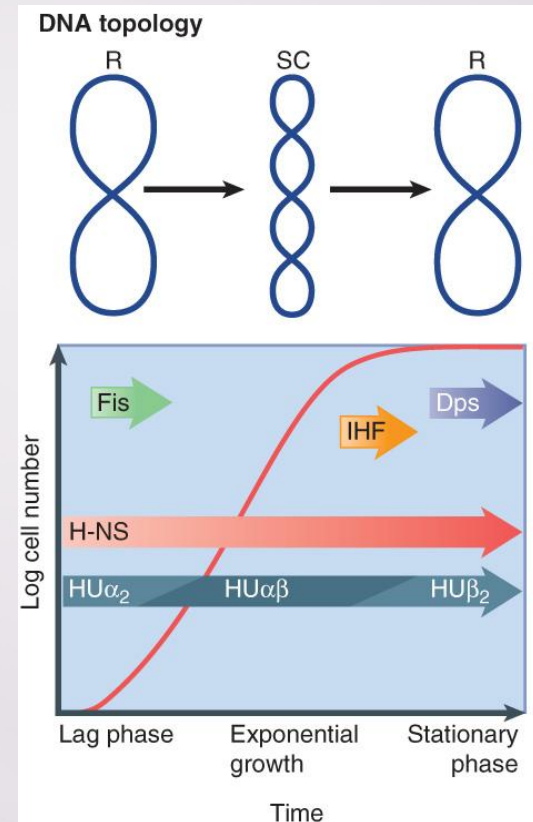
Figure B: The nucleoid spills out of a lysed *E. coli* cell in the form of loops of a fiber.

A) Photo courtesy of the Molecular and Cell Biology Instructional Laboratory Program, University of California, Berkeley.

3.3 The Bacterial Genome Is a Nucleoid with Dynamic Structural Properties

- Bacterial condensin complexes (SMC-ScpAB or MukBEF) function in chromosome structure and segregation.

Figure : Growth phase and elements that affect nucleoid structure. A typical growth curve for *E. coli* growing in batch culture begins with a lag phase followed by the log phase of exponential growth and, finally, stationary phase (when the cells stop growing).



3.4 The Bacterial Genome Is Supercoiled and Has Four Macrod domains

- The circular bacterial genome has four macrodomains (*ori*, right, *ter*, left) that adopt replication-associated spatial patterns.
- The average density of supercoiling is approximately 1 turn/100 bp.
- The nucleoid has about 400 independent negatively supercoiled **domains**.

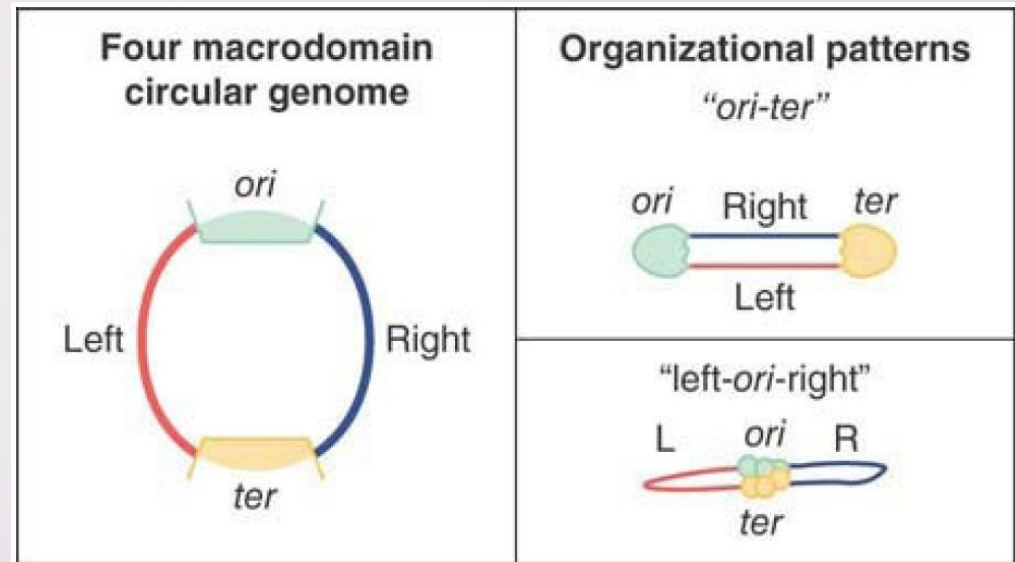
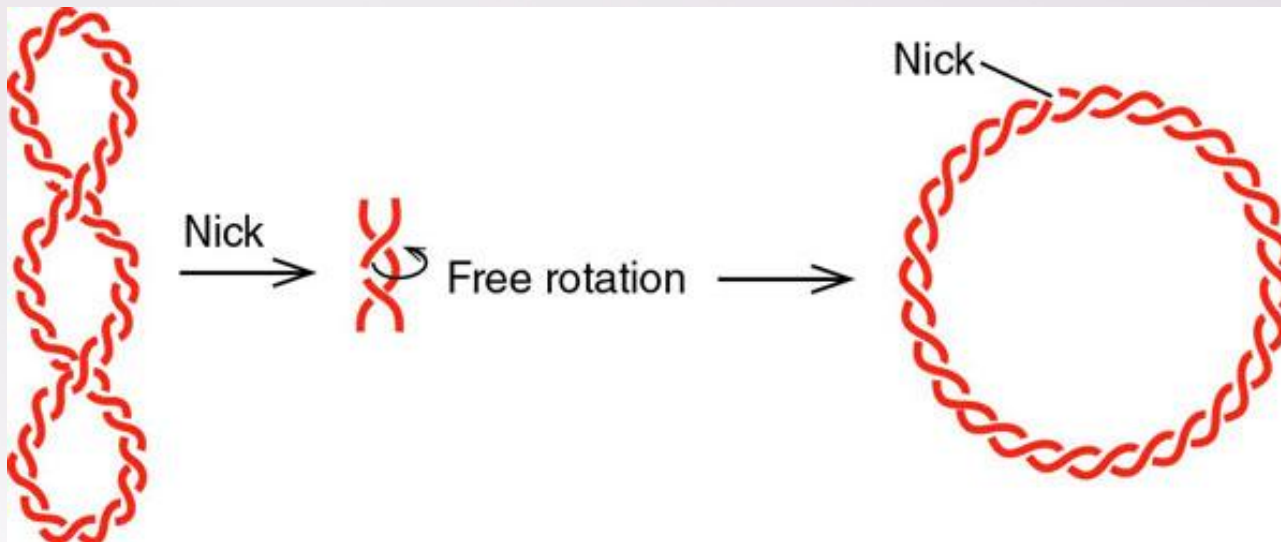


FIGURE : Large-scale organizational patterns of the macrodomains in bacteria. The domains are delimited by the origin (*ori*) and termination (*ter*) regions, creating two different replichores termed left and right.

Supercoiling can be constrained if proteins are bound to the DNA to hold it in a particular three-dimensional configuration.



If a supercoiled DNA is free, its path is *unconstrained*, and negative supercoils generate a state of torsional tension that is transmitted freely along the DNA within a domain. Torsional tension resulting from negative supercoils can be relieved by unwinding the double helix.

3.5 The Eukaryotic Chromosome and Chromatin

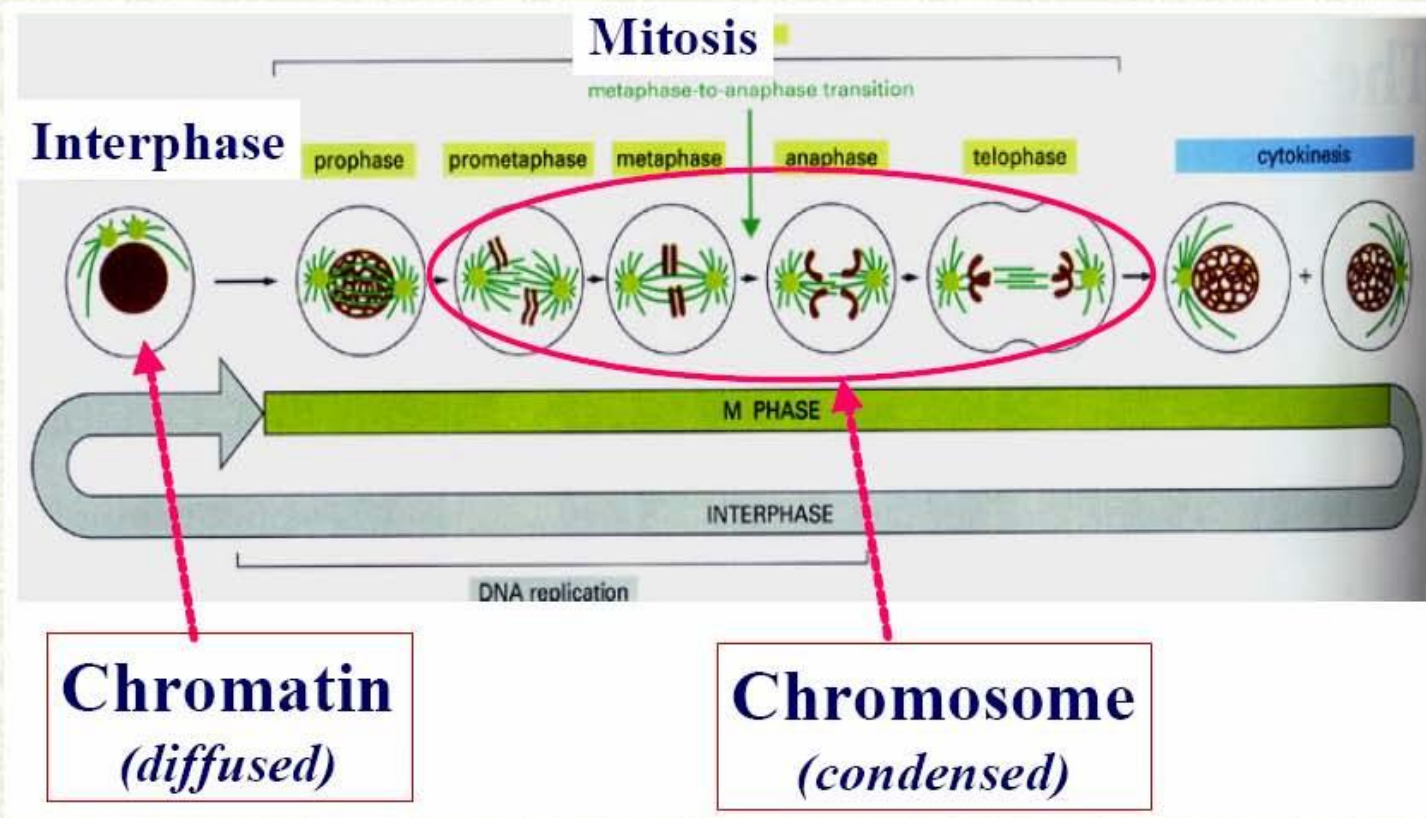
- **chromatin** – The state of nuclear DNA and its associated proteins during the interphase (between mitoses) of the eukaryotic cell cycle.

KEY CONCEPTS:

DNA of interphase chromatin is negatively supercoiled into independent domains averaging 85 kb.

Metaphase chromosomes have a protein scaffold to which the loops of supercoiled DNA are attached.

In the first phase of mitosis the **chromatin** fibers condense into **chromosomes**.



The **nucleus** of a nondividing cell (**interphase**) is filled with **diffuse** material called **Chromatin**, so called because early microscopists found that it stained brightly with certain dyes.

What controls the distribution of interphase chromatin within the nucleus?

Interphase chromatin is a tangled-appearing mass occupying a large part of the nuclear volume.

This is in contrast with the highly organized and reproducible ultrastructure of mitotic chromosomes.



Figure : Chromosomes occupy chromosome territories in the nucleus and are not entangled with one another.

3.6 Specific Sequences Attach DNA to an Interphase Matrix

- DNA is attached to the **nuclear matrix** at specific sequences called **matrix attachment regions (MARs)**.
- The MARs are A-T-rich but do not have any specific consensus sequence.

3.7 Chromatin Is Divided into Euchromatin and Heterochromatin

- We can see individual chromosomes only during mitosis.
- During interphase, the general mass of chromatin is in the form of **euchromatin**, which is slightly less tightly packed than mitotic chromosomes.

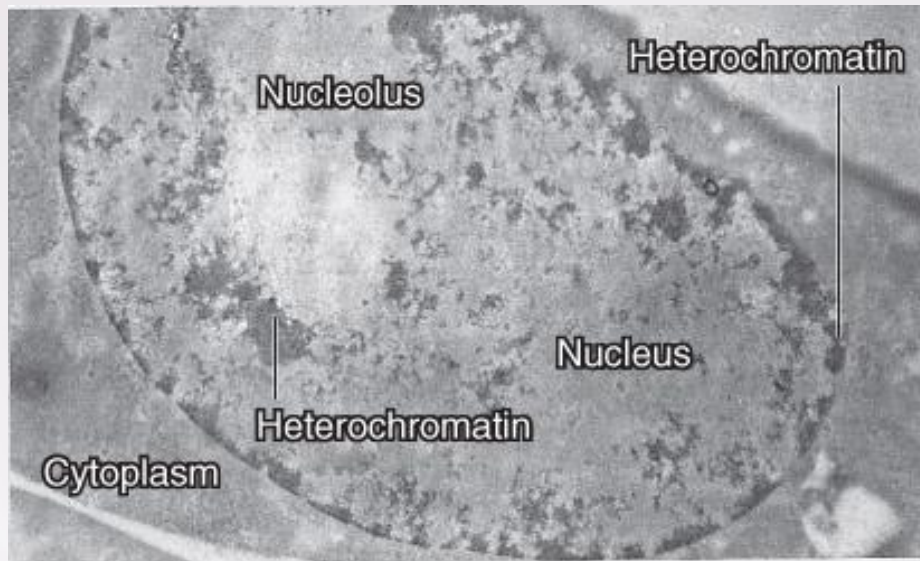


Figure : A thin section through a nucleus stained with Feulgen.

The common features of constitutive heterochromatin :

- It is permanently or nearly always condensed.
- It replicates late in S phase
- It has a reduced frequency of genetic recombination relative to euchromatic gene-rich areas of the genome.
- It often consists of multiple repeats of a few sequences of DNA
- The DNA in this region are not transcribed or are transcribed at very low levels.
- The density of genes in this region is very much reduced compared with euchromatin.

3.8 The Eukaryotic Chromosome Is a Segregation Device

- A eukaryotic chromosome is held on the mitotic **spindle** by the attachment of microtubules to the **kinetochore** that forms in its centromeric region.
- **microtubule organizing center (MTOC)** – A region from which microtubules emanate.
 - In animal cells the centrosome is the major microtubule organizing center.

3.8 The Eukaryotic Chromosome Is a Segregation Device

- **centromere** – A constricted region of a chromosome that includes the site of attachment (the kinetochore) to the mitotic or meiotic spindle.
 - It consists of unique DNA sequences and proteins not found anywhere else in the chromosome.
- **acentric fragment** – A fragment of a chromosome (generated by breakage) that lacks a centromere and is lost at cell division.

3.8 The Eukaryotic Chromosome Is a Segregation Device

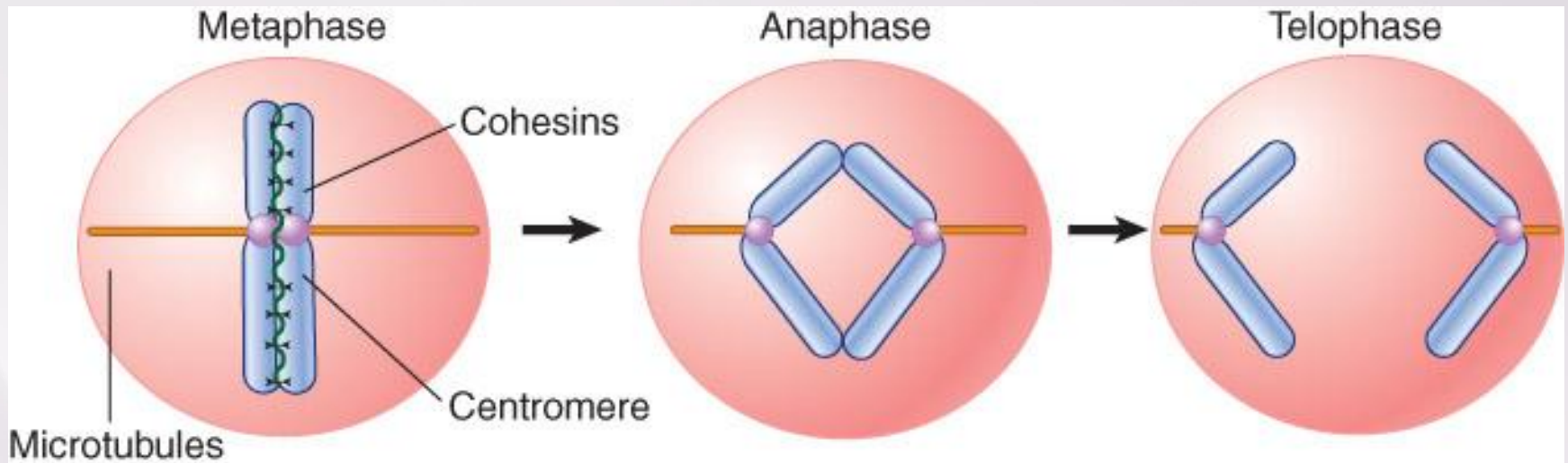
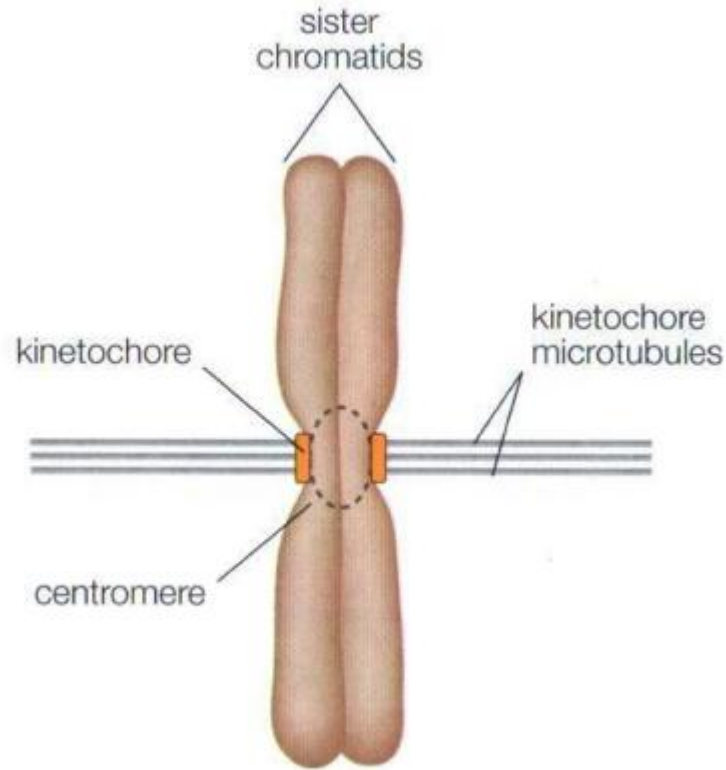
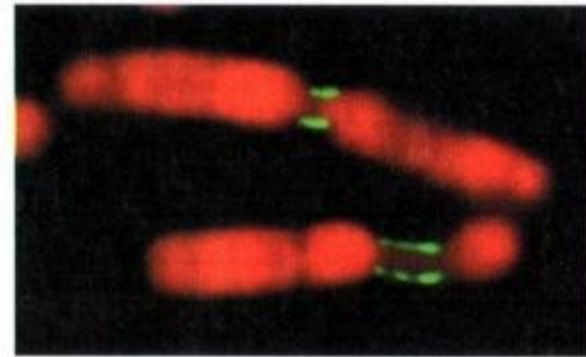


Figure : Chromosomes are pulled to the poles via microtubules that attach at the centromeres.



(a)



(b)

At the centromere (着丝粒), each chromatid develops a special protein complex called a kinetochores (着丝点) that lock onto the ends of microtubule

3.9 Regional Centromeres Contain a Centromeric Histone H3 Variant and Repetitive DNA

- Installation of the centromere-specific histone H3 is an epigenetic and primary determinant that specifies a functional centromere.
- Centromeres in higher eukaryotic chromosomes contain large amounts of repetitive DNA and unique histone H3 variants.
- The function of the ever-present repetitive centromeric DNA is not known.

3.9 Regional Centromeres Contain a Centromeric Histone H3 Variant and Repetitive DNA

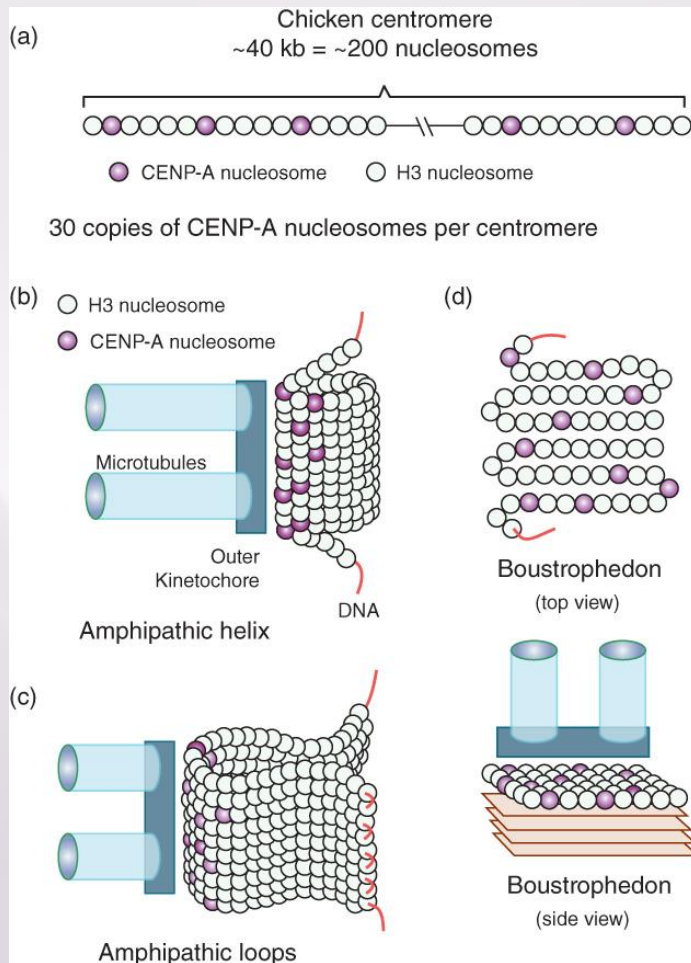


Figure : Organization of CENP-A and H3 Nucleosomes in Centromeres.

(a) Centromeres are ~40 kb long in chicken, corresponding to 200 nucleosomes per centromere. Of these, 30 are predicted to contain CENP-A (roughly 1 in 6–8 centromeric nucleosomes). Thus, centromeric chromatin is largely composed of nucleosomes containing histone H3. **(b and c)** The CENP-A chromatin was originally suggested to form an amphipathic organization, with CENP-A on the exterior facing the kinetochore, and H3 largely on the interior. This chromatin was proposed to form either a helix or loop structure. **(d)** The boustrophedon model of centromeric CENP-A-containing chromatin was proposed based on super-resolution microscopy.

3.10 Telomeres Have Simple Repeating Sequences

- The **telomere** is required for the stability of the chromosome end.
- A telomere consists of a simple repeat where a G-rich strand at the 3' terminus typically has a sequence of $(T/A)_{1-4} G_{>2}$.



Figure : A typical telomere has a simple repeating structure with a G-T-rich strand that extends beyond the C-A-rich strand.

3.11 Telomeres Seal the Chromosome Ends and Function in Meiotic Chromosome Pairing

- The protein TRF2 catalyzes a reaction in which the 3' repeating unit of the G+T-rich strand forms a loop by displacing its homolog in an upstream region of the telomere.
- The absence of any free end might be the crucial feature that stabilizes the chromosome.

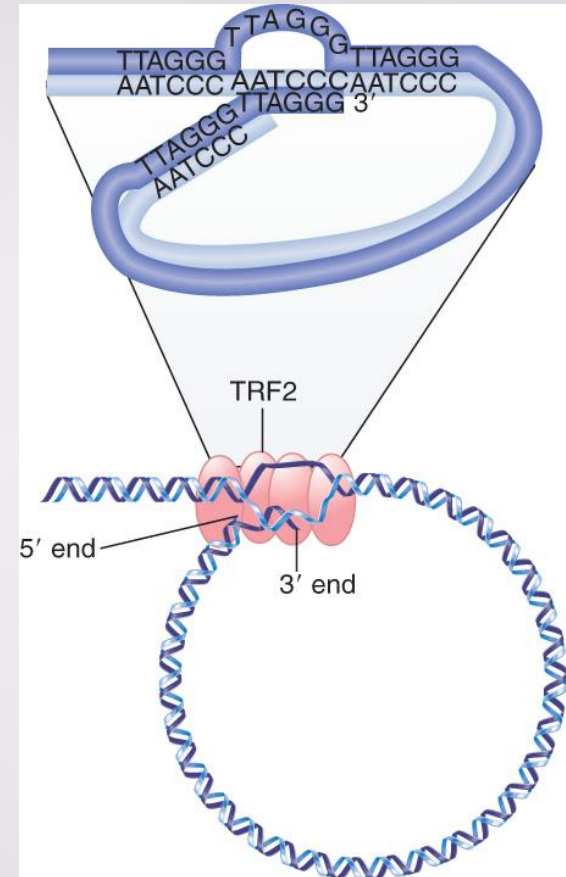
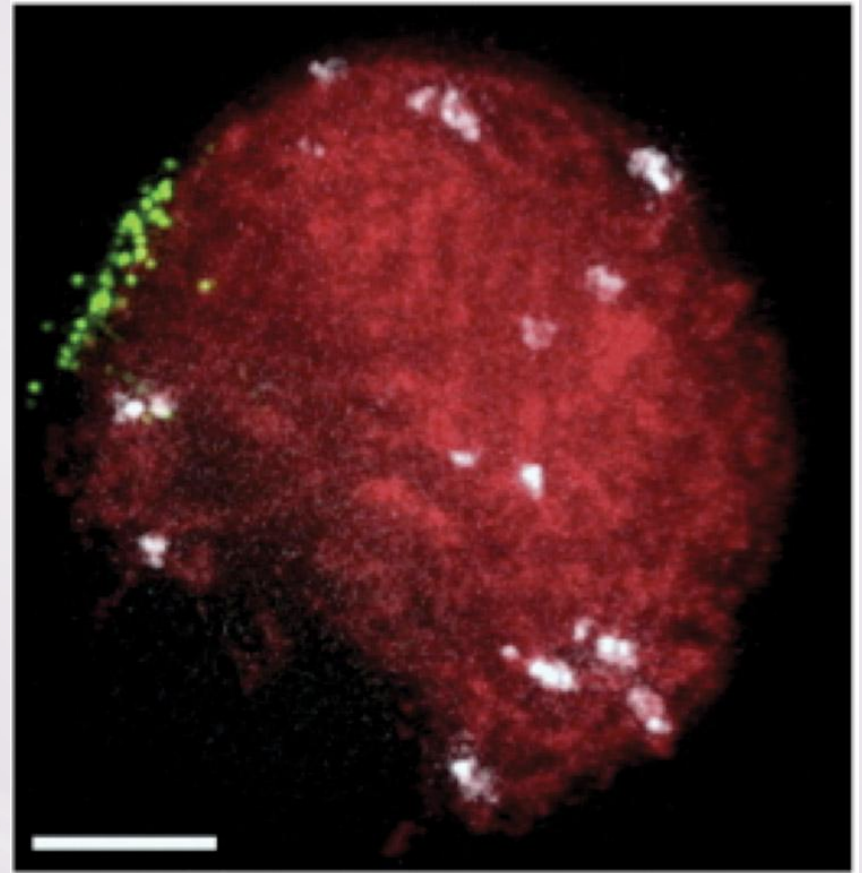


Figure : A loop forms at the end of chromosomal DNA.

3.11 Telomeres Seal the Chromosome Ends and Function in Meiotic Chromosome Pairing

- Telomeres promote pairing, synapsis, and recombination during meiosis via links to the cytoskeleton through nuclear envelope proteins.

Figure : The meiotic telomere cluster is visualized by telomere FISH.



3.12 Telomeres Are Synthesized by a Ribonucleoprotein Enzyme

- **Telomerase** uses the 3'–OH of the G+T telomeric strand and its own RNA template to iteratively add tandem repeats (5'-TTAGGG-3' in human) to the 3' end at each chromosomal terminus.
- Telomerase uses a reverse transcriptase to extend the very ends of the chromosomes and solve the so-called end **replication problem**.

3.12 Telomeres Are Synthesized by a Ribonucleoprotein Enzyme

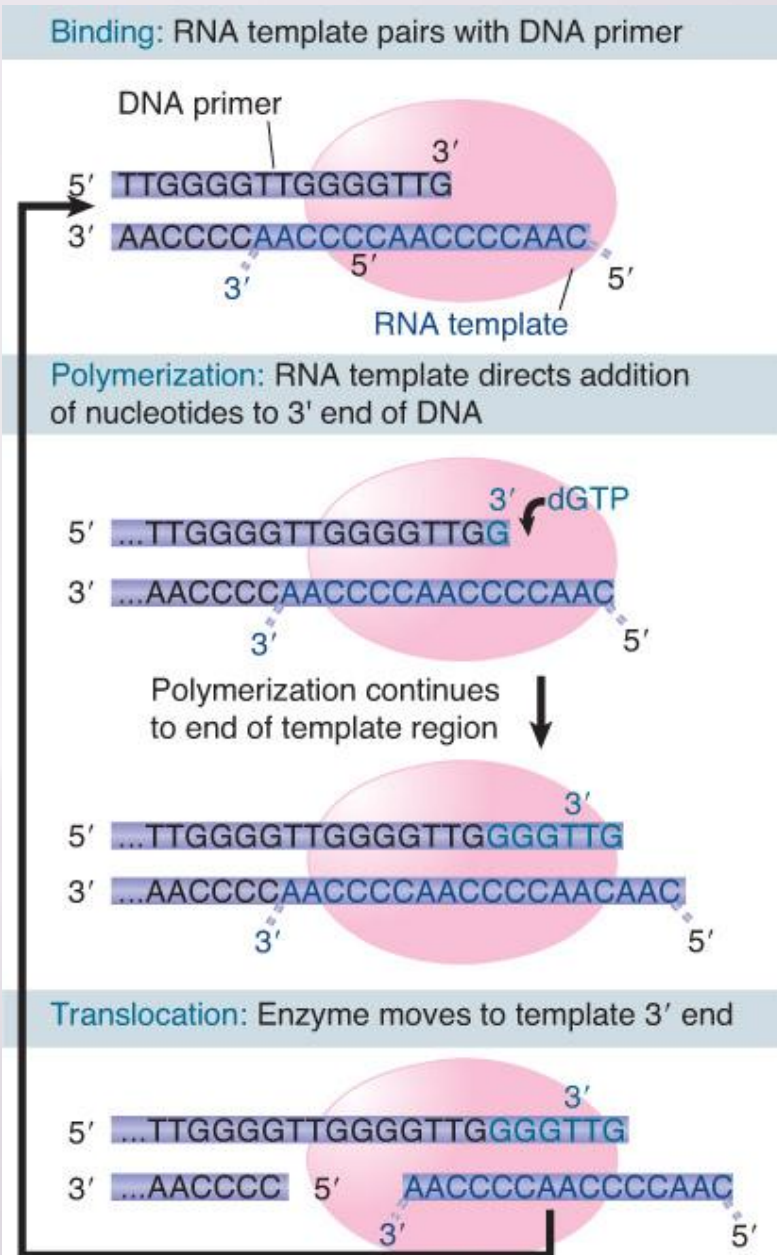


Figure : Telomerase positions itself by base pairing between the RNA template and the protruding single-stranded DNA primer.

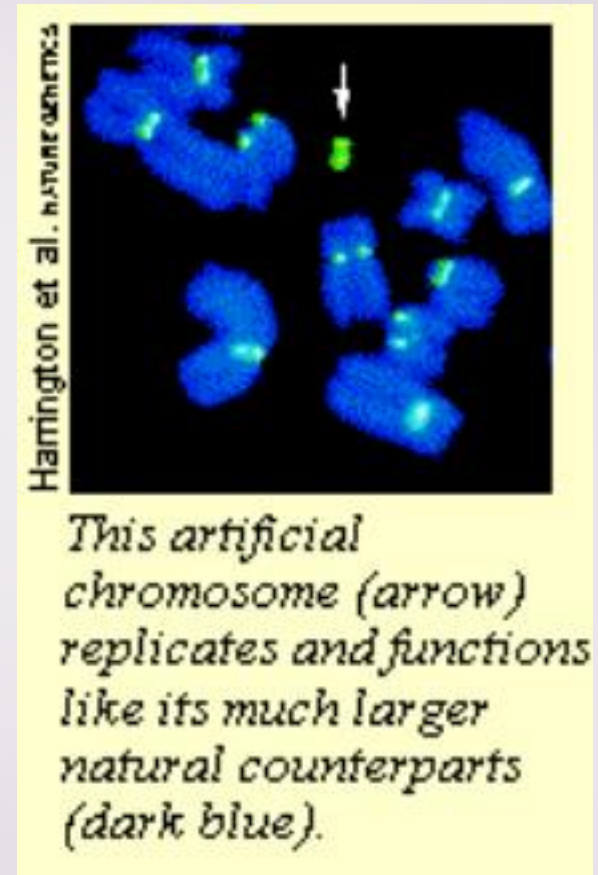
The functions of telomere :

- The first is to protect the chromosome end and maintain its stability. —for example, the end generated by a double-strand break—becomes a target for repair systems. The cell must be able to distinguish the telomere and any other end.
- The second is to allow the telomere to be extended. If it is not extended, it becomes shorter with each replication cycle (because replication cannot initiate at the very end).
- The third is to facilitate meiotic chromosome reorganization for efficient pairing and recombination of homologous chromosomes.

The minimum features required for existence as a chromosome are as follows:

- Telomeres to ensure survival
- A centromere to support segregation
- An origin to initiate replication

1997, First Human Artificial Chromosome



3.13 DNA Is Organized in Arrays of Nucleosomes

Introduction

- **nucleosome** – The basic structural subunit of chromatin, consisting of about 200 bp of DNA and an octamer of **histone** proteins.
- **histone tails** – Flexible amino- or carboxy-terminal regions of the core histones that extend beyond the surface of the nucleosome.
 - Histone tails are sites of extensive posttranslational modification.

3.13 DNA Is Organized in Arrays of Nucleosomes

Introduction

- **10 nm fiber** – A linear array of nucleosomes generated by unfolding from the natural condition of chromatin.
- **linker histones** – A family of histones (such as histone H1) that are not components of the nucleosome core.
 - Linker histones bind nucleosomes and/or linker DNA and promote 30 nm fiber formation.

3.13 DNA Is Organized in Arrays of Nucleosomes

Introduction

- **30 nm fiber** – A coil of nucleosomes.
 - It is the basic level of organization of nucleosomes in chromatin.
- **nonhistone** – Any structural protein found in a chromosome except one of the histones.

3.13 DNA Is Organized in Arrays of Nucleosomes

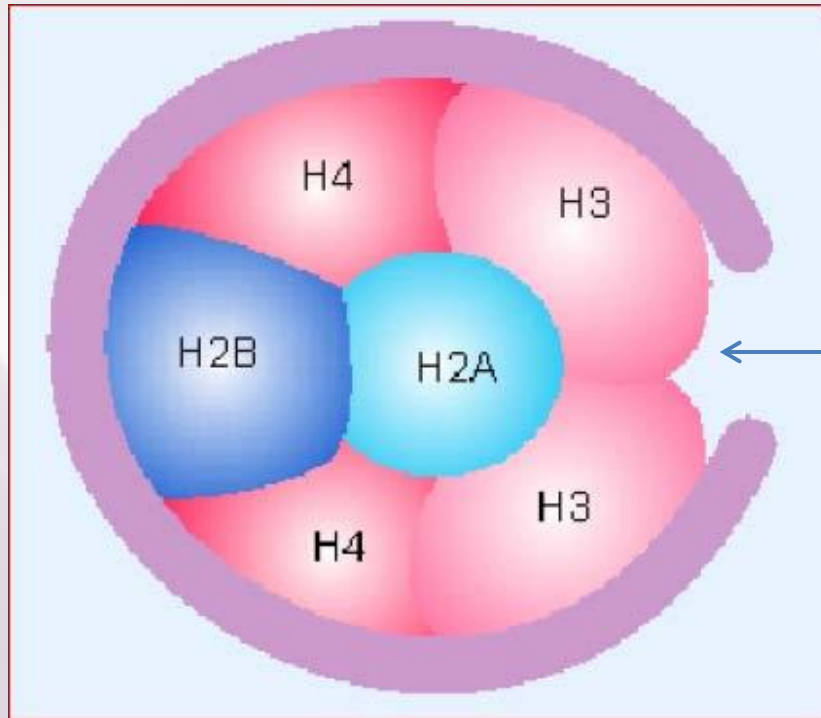
- **MNase** (micrococcal nuclease) cleaves linker DNA and releases individual nucleosomes from chromatin.
- More than 95% of the DNA is recovered in nucleosomes or multimers when MNase cleaves DNA in chromatin.
- The length of DNA per nucleosome varies for individual tissues or species in a range from 154 to 260 bp.
- Nucleosomal DNA is divided into the core DNA and linker DNA depending on its susceptibility to MNase.

3.13 DNA Is Organized in Arrays of Nucleosomes

- A nucleosome contains approximately 200 bp of DNA and two copies of each core histone (H2A, H2B, H3, and H4).
- DNA is wrapped around the outside surface of the protein octamer.
- The **histone octamer** has a structure of an H3₂-H4₂ tetramer associated with two H2A-H2B dimers.

Nucleosome: DNA combines with histones

组蛋白是染色体的结构蛋白：
有**H1**、**H2A**、**H2B**、**H3**及**H4**五种，
与**DNA**共同组成核小体。



核小体核心是由**H2A**、**H2B**、**H3**、**H4**各两分子生成的八聚体

3.13 DNA Is Organized in Arrays of Nucleosomes

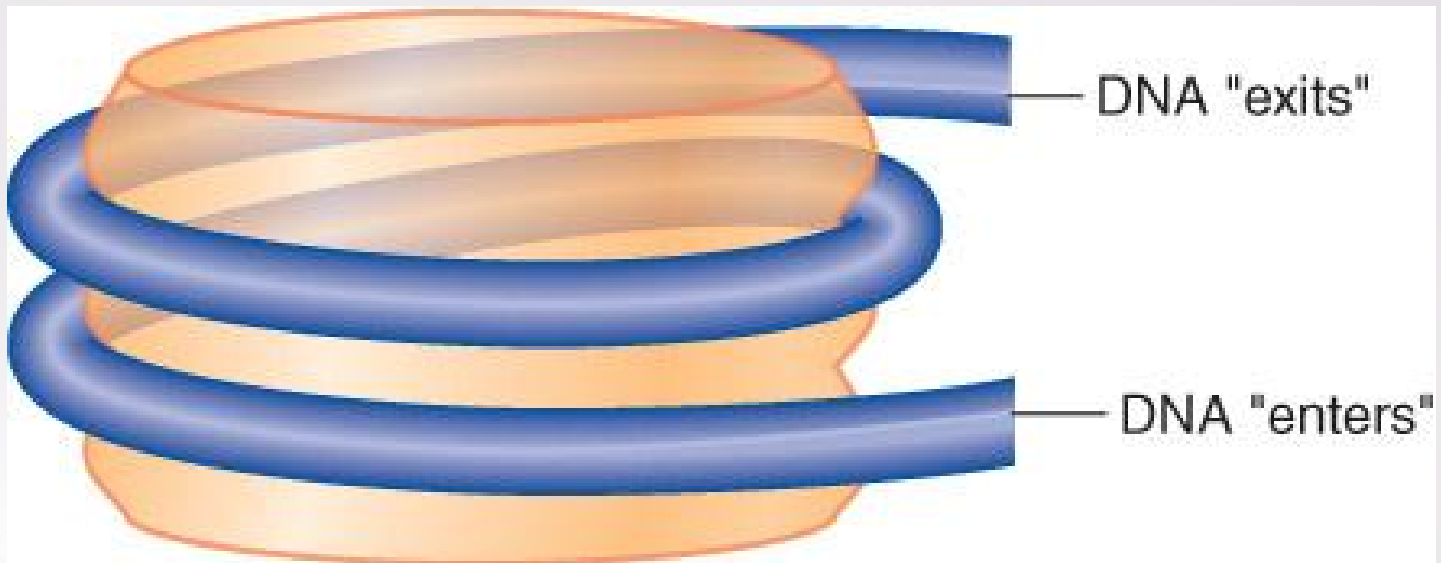


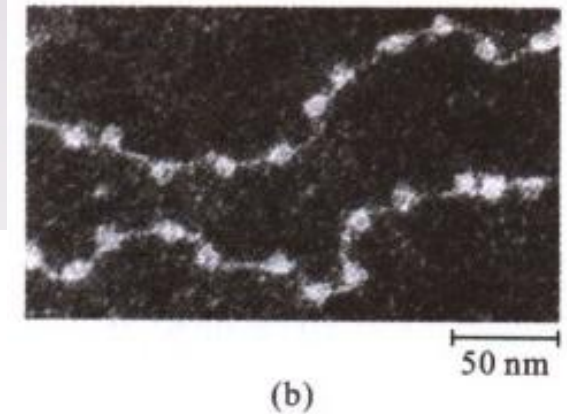
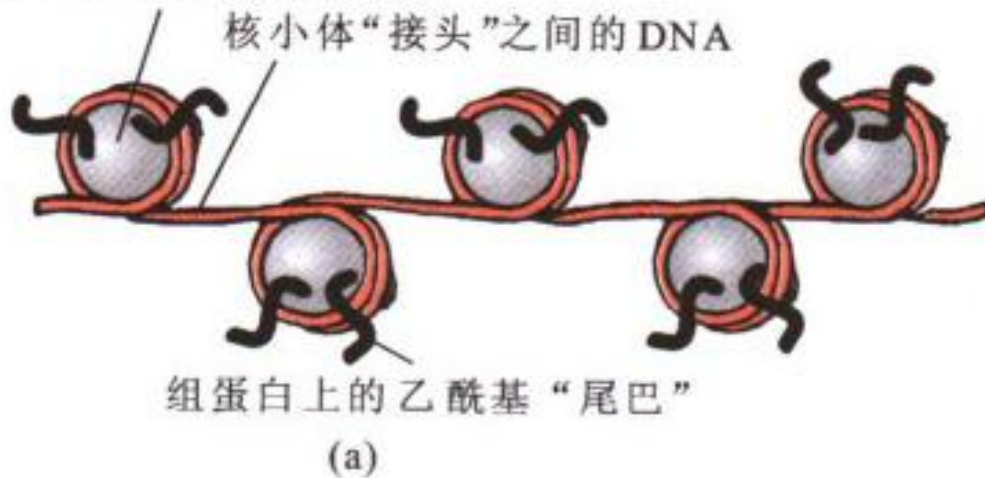
Figure : The nucleosome is a cylinder with DNA organized into ~one and two-thirds turns around the surface.

The structure of nucleosome

核小体结构示意图

Nucleosome (核小体) 是染色质的基本结构单位，由~200 bp DNA和组蛋白八聚体组成

核小体中的组蛋白核心结构

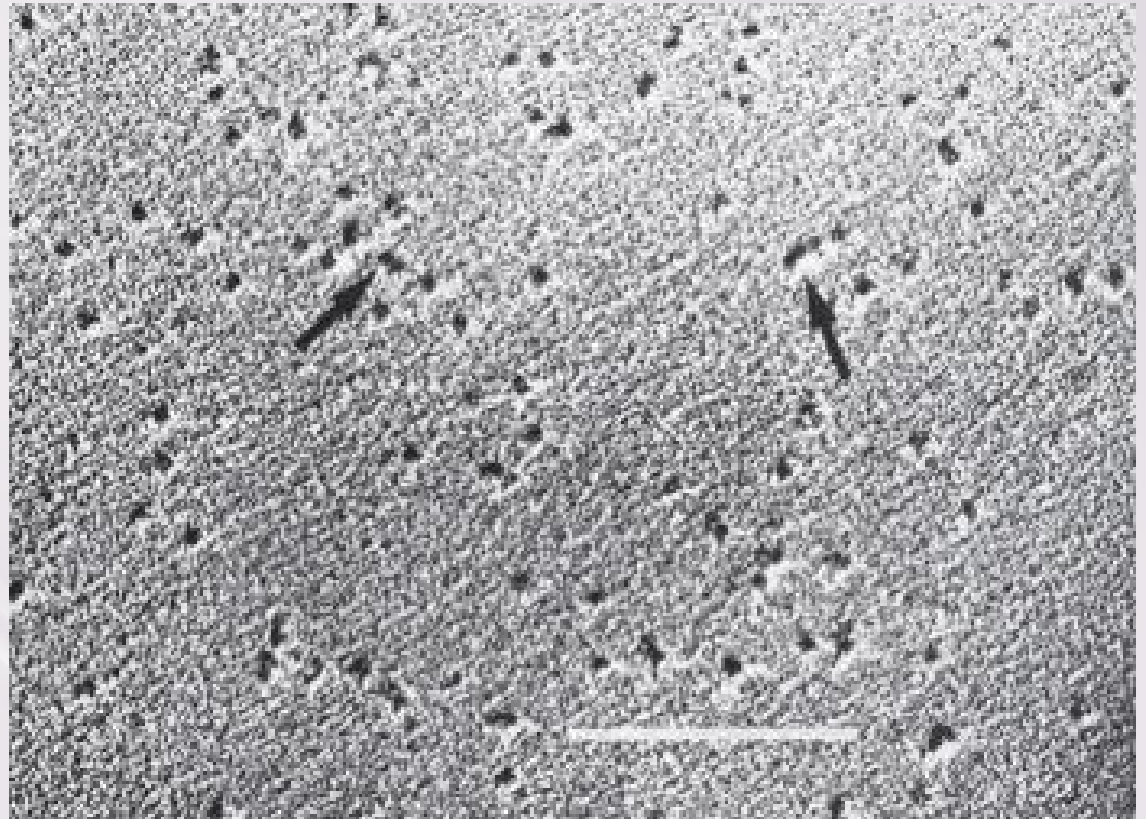


电镜下看到的染色质纤维细丝，是由DNA和组蛋白组成的许多核小体连成的念珠状结构

由许多核小体构成了连续的染色质DNA细丝。

3.13 DNA Is Organized in Arrays of Nucleosomes

Figure : Individual nucleosomes are released by digestion of chromatin with **micrococcal nuclease**. The bar is 100 nm.



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染色质中DNA与蛋白质紧密结合的实验证据：

- 1、染色质DNA的T_m值比自由DNA高，说明在染色质中DNA极可能与蛋白质分子相互作用。
- 2、在染色质状态下，由DNA聚合酶和RNA聚合酶催化的DNA复制和转录活性大大低于在自由DNA中的反应。
- 3、DNA酶I（DNaseI）对染色质DNA的消化远远慢于对纯DNA的作用。
- 4、用微球菌核酸酶处理染色质以后进行电泳，便可以得到一系列片段，这些被保留的DNA片段均为200bp基本单位的倍数。

3.13 DNA Is Organized in Arrays of Nucleosomes

- The **core DNA** is the length of 145–147 bp that is found on the core particles produced by prolonged digestion with MNase.
- **Linker DNA** is the region of 7 to 115 bp that is susceptible to early cleavage by nucleases.

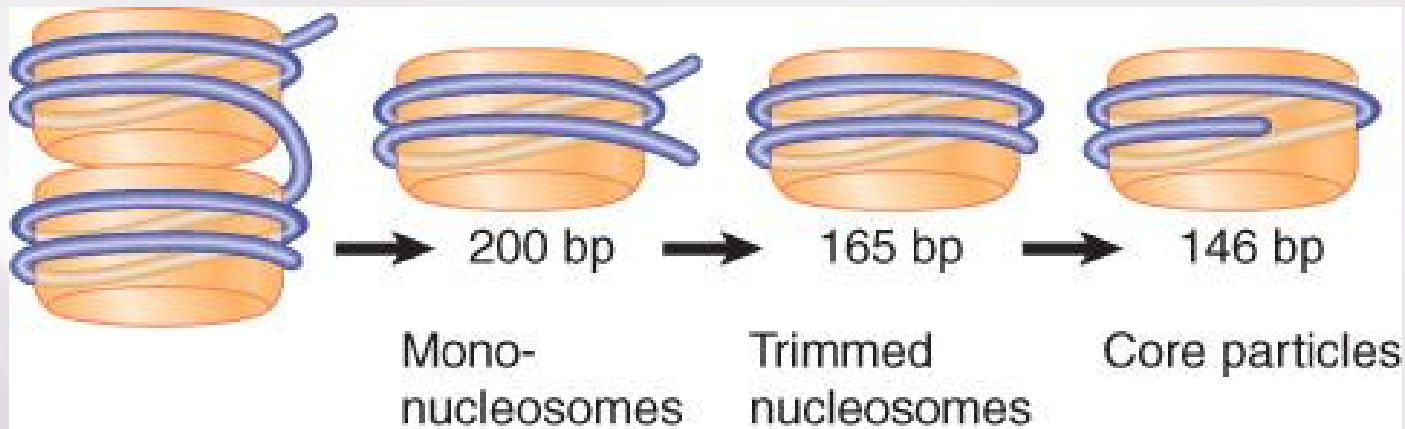


Figure : Micrococcal nuclease initially cleaves between nucleosomes.

3.14 The Nucleosome Is the Subunit of All Chromatin

- Each histone is extensively interdigitated with its partner.
- All **core histones** have the structural motif of the **histone fold**. N- and C-terminal histone tails extend out of the nucleosome.
- H1 is associated with linker DNA and can lie at the point where DNA enters or exits the nucleosome.

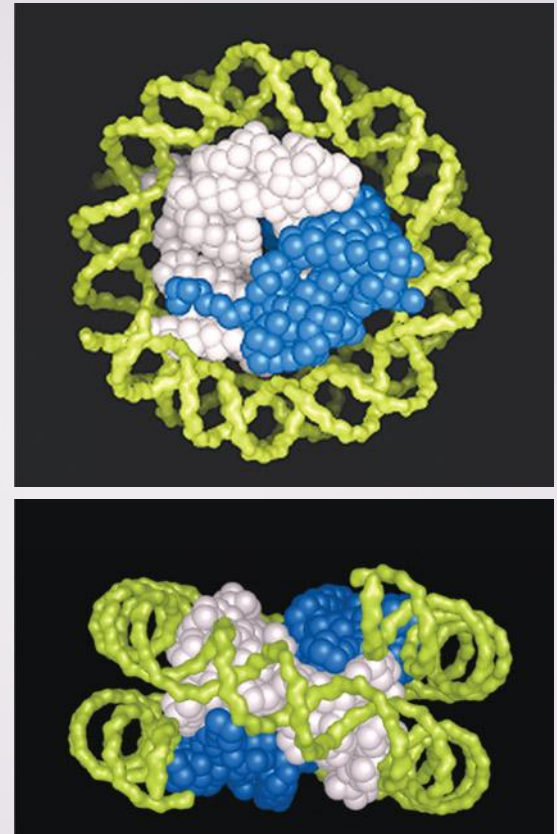


Figure : The crystal structure of the histone core octamer is represented in a space-filling model.

3.14 The Nucleosome Is the Subunit of All Chromatin



Figure : The histone fold (left) consists of two short α -helices flanking a longer α -helix. Histone pairs (H3 + H4 and H2A + H2B) interact to form histone dimers (right).

Structures from Protein Data Bank 1HIO.

G. Arents, et al., *Proc. Natl. Acad. Sci.*

USA 88 (1991): 10145–10152.

3.14 The Nucleosome Is the Subunit of All Chromatin

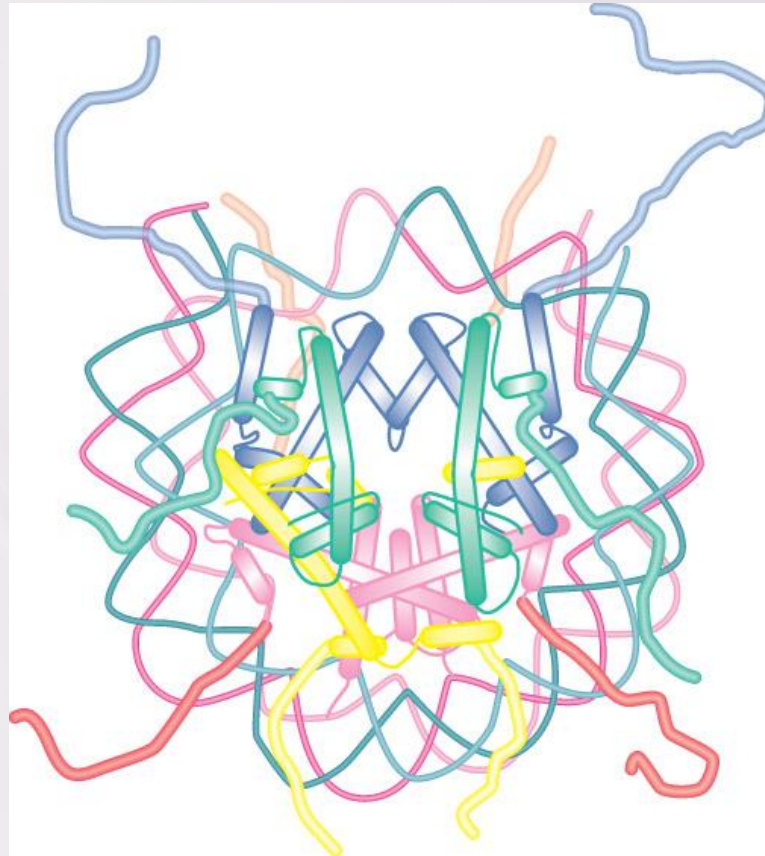
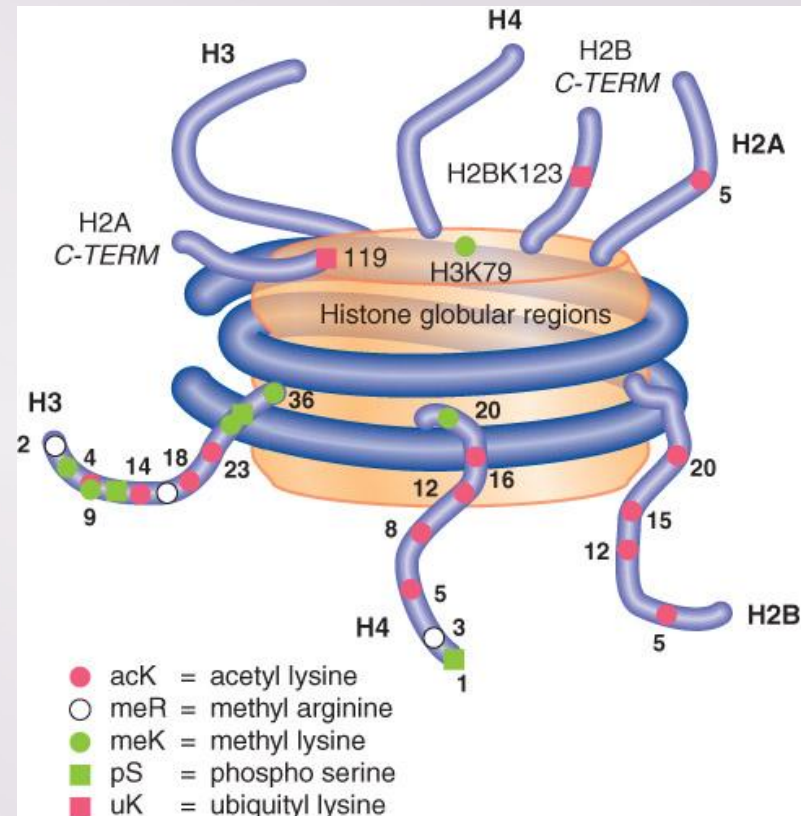


Figure : The histone fold domains of the histones are located in the core of the nucleosome.

3.15 Nucleosomes Are Covalently Modified

- Histones are modified by **methylation**, **acetylation**, **phosphorylation**, **ubiquitylation**, **sumoylation**, **ADP-ribosylation**, and other modifications.
- Combinations of specific histone modifications define the function of local regions of chromatin; this is known as the **histone code** hypothesis.



3.15 Nucleosomes Are Covalently Modified

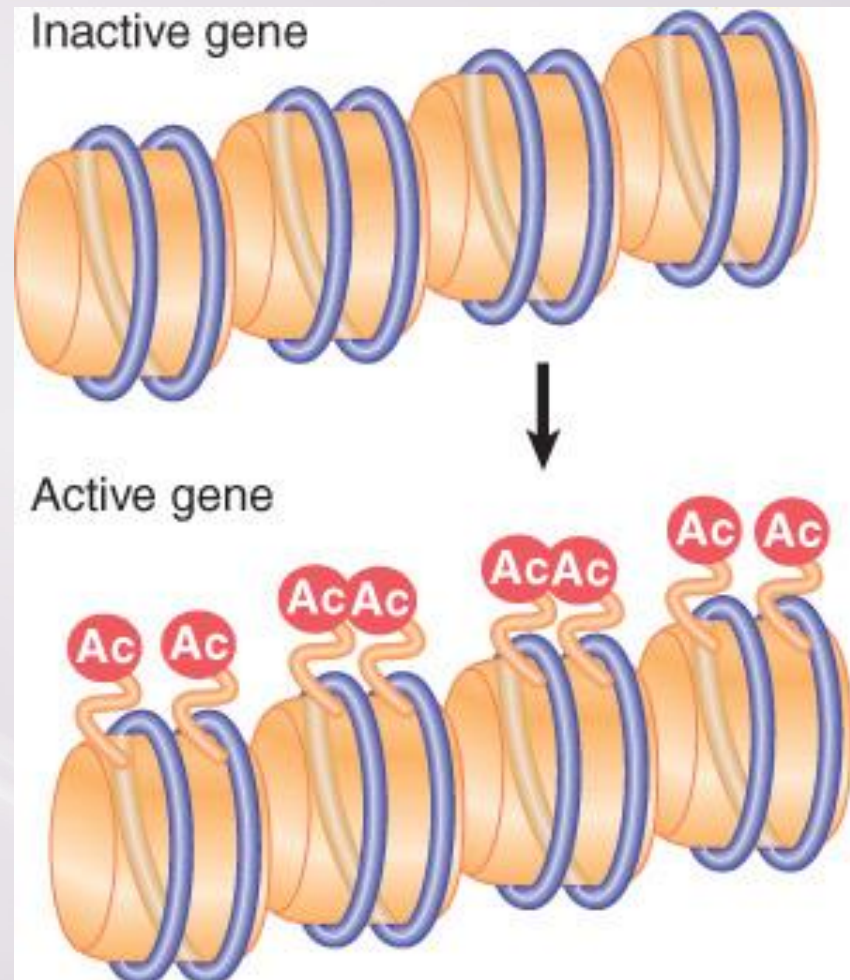


Figure : Acetylation associated with gene activation.

3.16 Histone Variants Produce Alternative Nucleosomes

- All core histones except H4 are members of families of related variants.
- **Histone variants** can be closely related to or highly divergent from canonical histones.
- Different variants serve different functions in the cell.

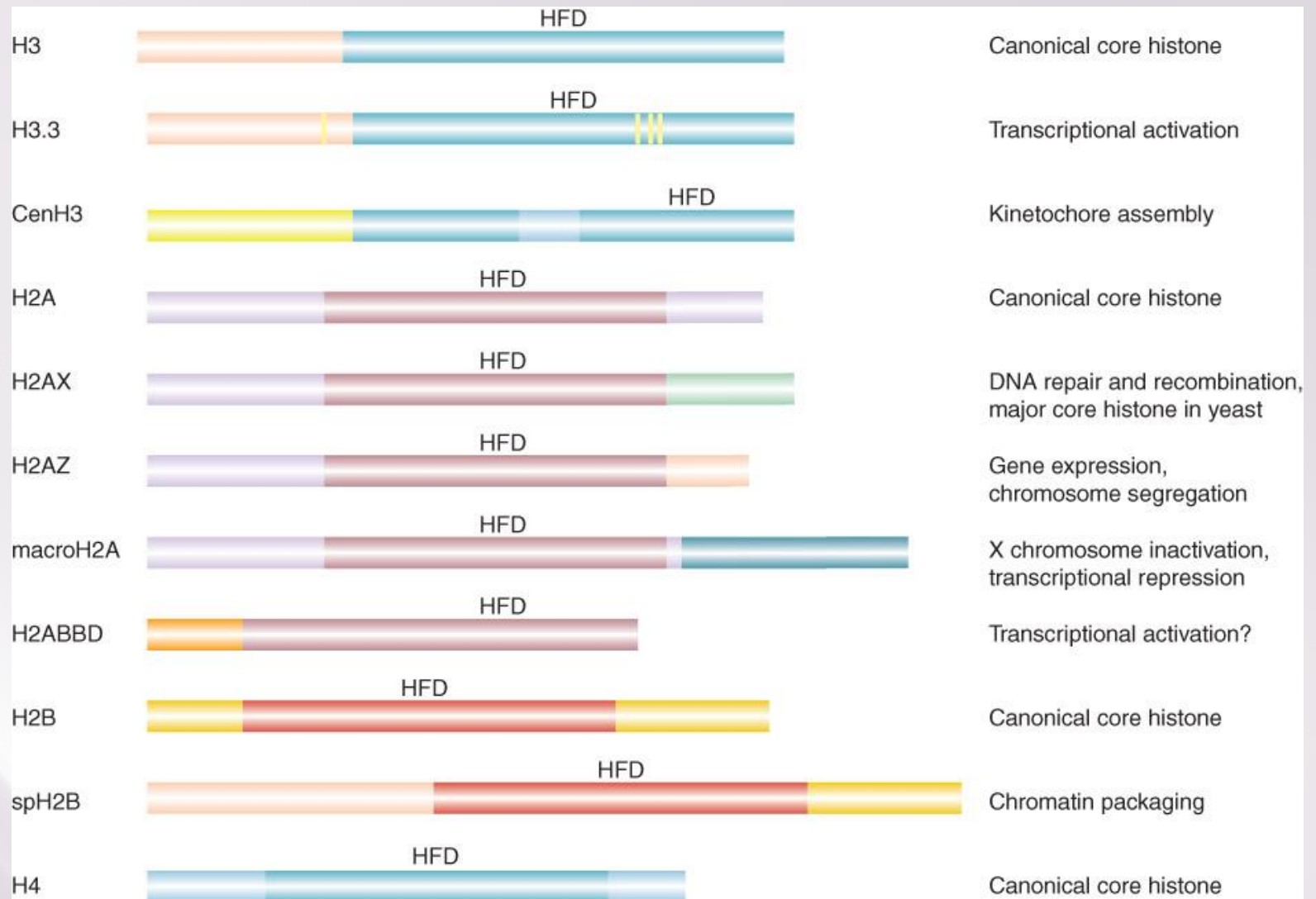
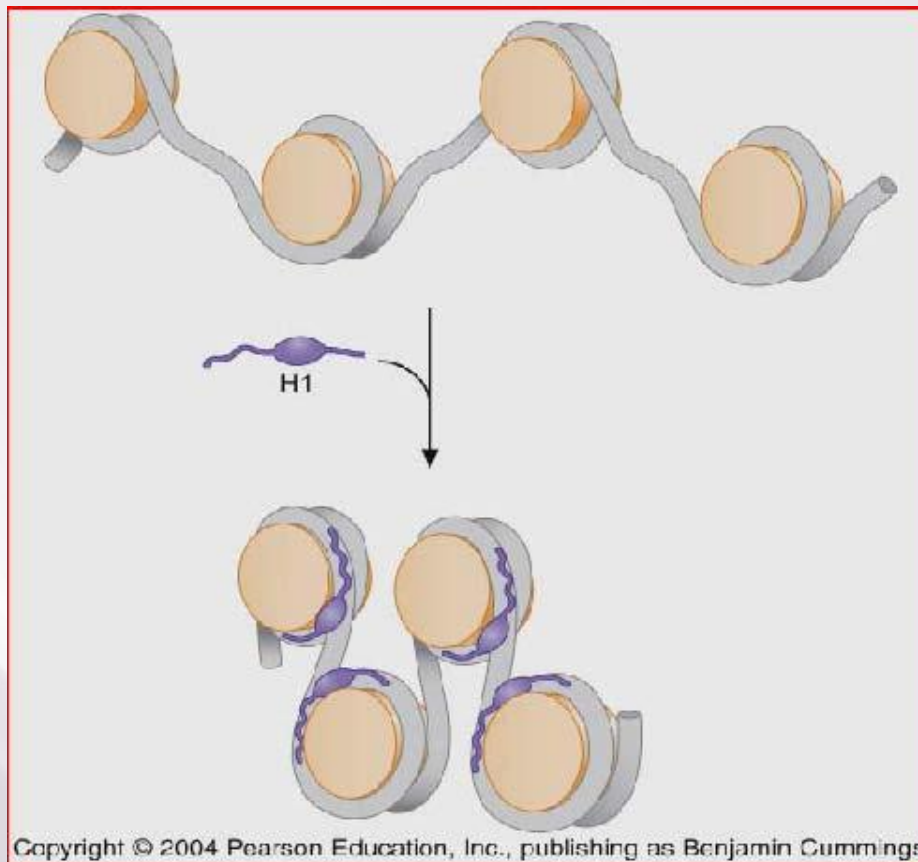


Figure : The major core histones contain a conserved histone-fold domain.

3.17 The Path of Nucleosomes to Chromatin Fiber and Chromosome

How does the higher-order chromatin and chromosome structure form?

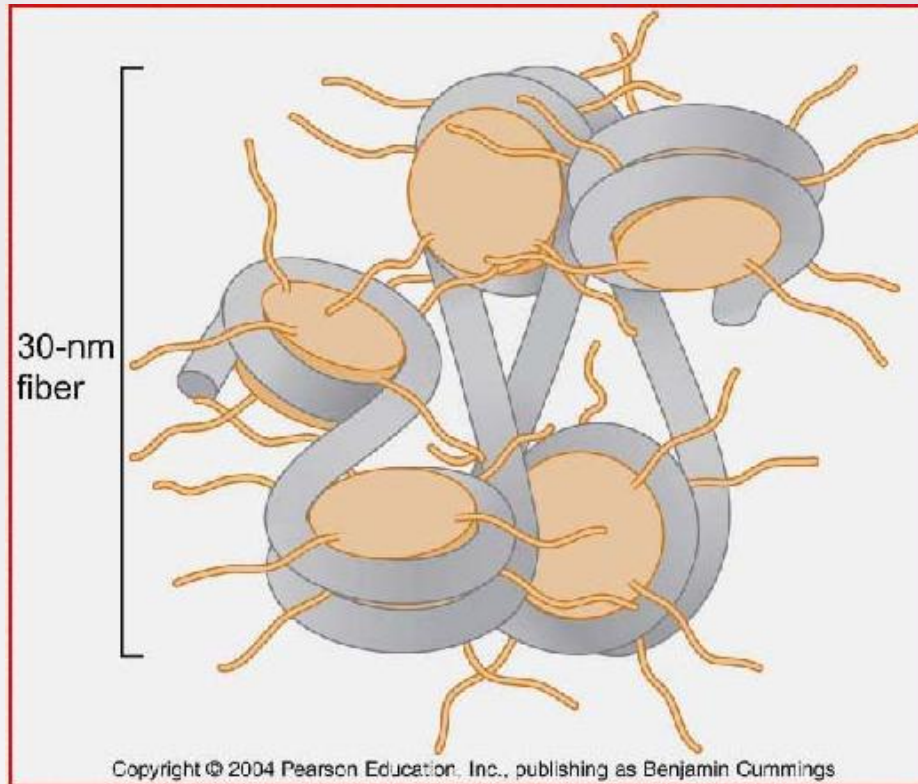
(1) Histone H1 binds to the linker DNA between nucleosome, inducing tighter DNA wrapping around the nucleosome



**The nucleosome
Filament**

染色质的10nm纤维细丝

(2) Nucleosome arrays can form more complex structures: the 30-nm fiber (“zigzag model”)

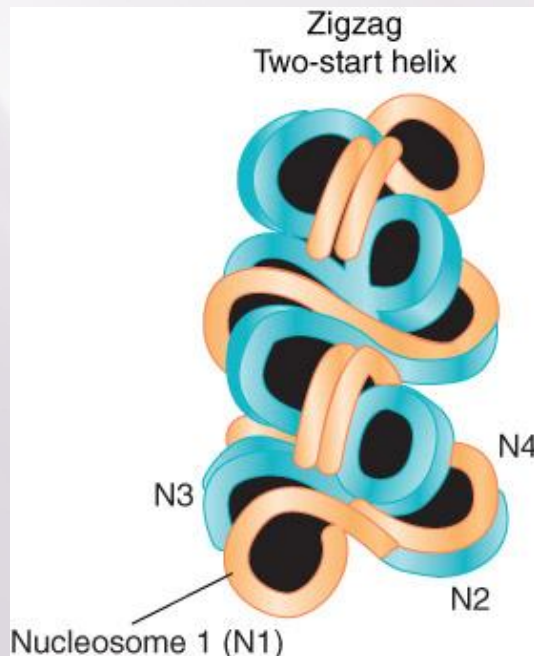
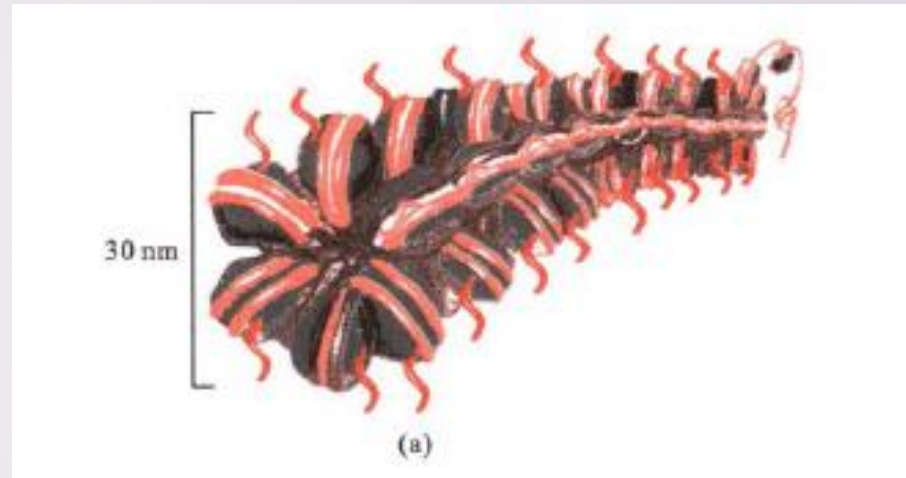
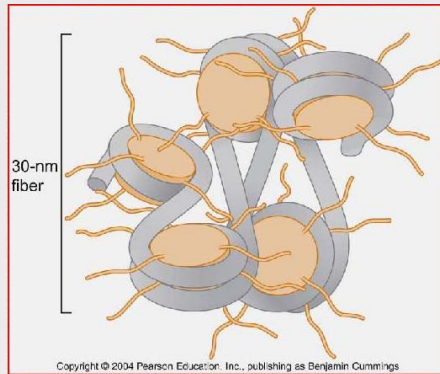


- 30 nm fibers have 6 nucleosomes/turn, organized into a solenoid.
- Histone H1 is required for formation of the 30 nm fiber.
- This fiber is the basic constituent of both interphase chromatin and mitotic chromosomes.

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(40-fold compaction)

30-nm fibers are a prevalent type of secondary structure that contain 6 nucleosomes/turn, organized into either a one-start solenoid or a two-start helix.



The 30-nm Fiber

染色质的30nm螺线管粗丝结构

Figure : The 30-nm fiber is a two-start helix consisting of two rows of nucleosomes coiled into a solenoid.

(3) Further compaction of DNA involves large loops of nucleosomal DNA

Additional 103-104-fold compaction is required, but the mechanism is unclear

The nuclear scaffold model is proposed

Eukaryotic DNA Has Loops and Domains Attached to a Scaffold

- Metaphase chromosomes have a protein **scaffold** to which the loops of supercoiled DNA are attached.
- **chromosome scaffold** – A proteinaceous structure in the shape of a sister chromatid pair, generated when chromosomes are depleted of histones.

Reprinted from *Cell*, vol. 12, J. R. Paulson and U. K. Laemmli, The structure of histone-depleted metaphase chromosomes, pp. 817-828. Copyright 1977, with permission from Elsevier [<http://www.sciencedirect.com/science/journal/00928674>]. Photo courtesy of Ulrich K. Laemmli, University of Geneva, Switzerland.

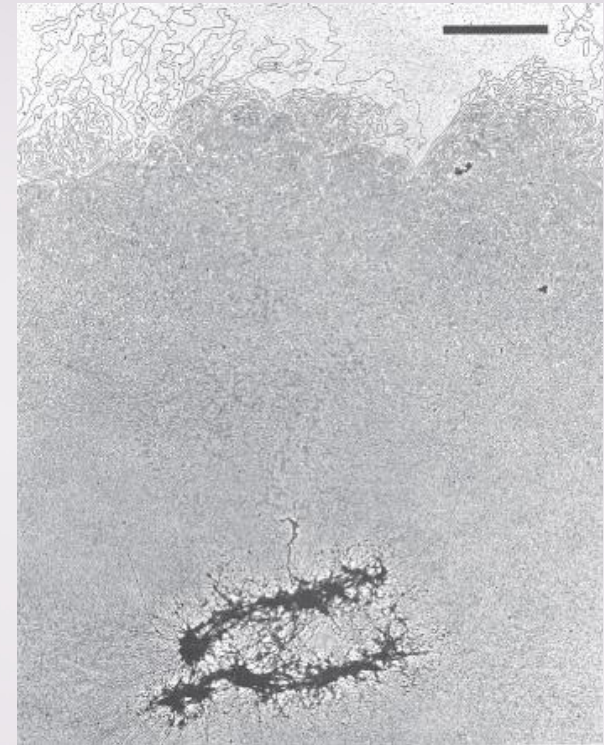
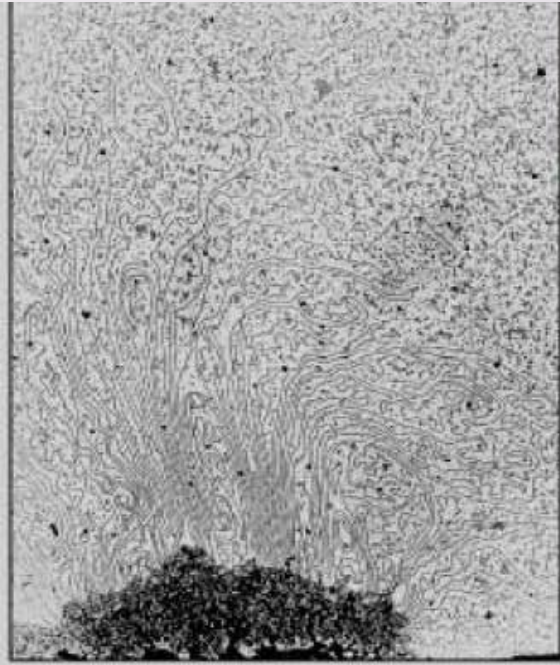
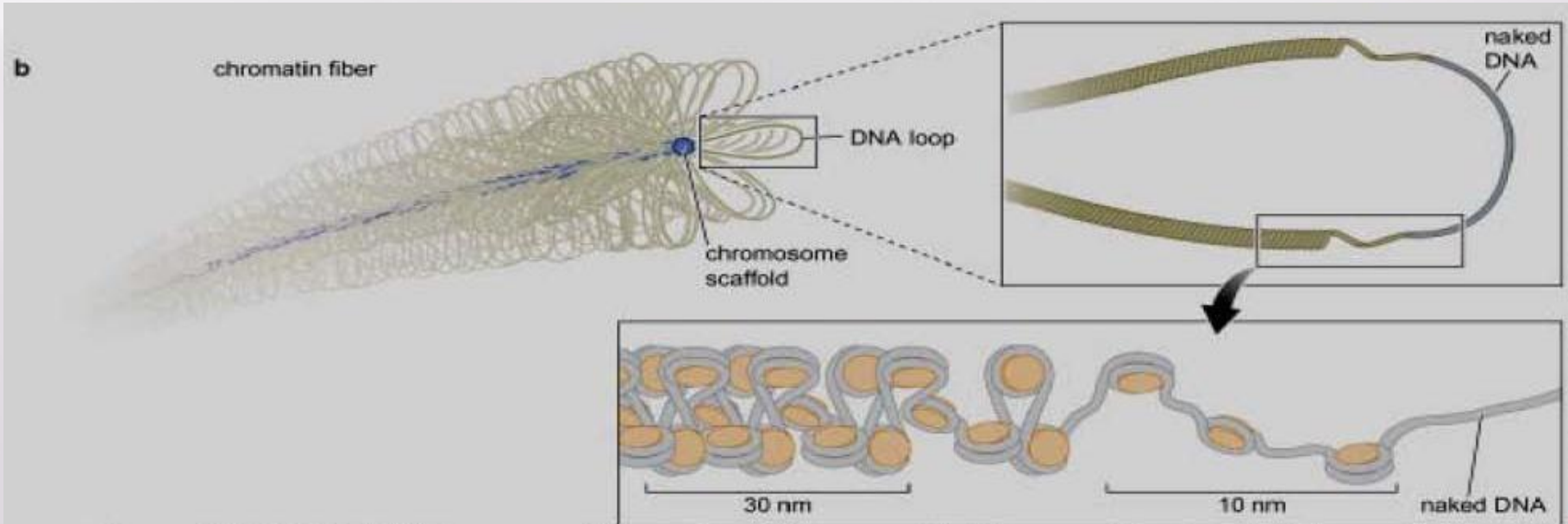


Figure : Histone-depleted chromosomes consist of a protein scaffold to which loops of DNA are anchored.



The nuclear scaffold model is proposed



From DNA to chromosome

Short region of DNA double helix

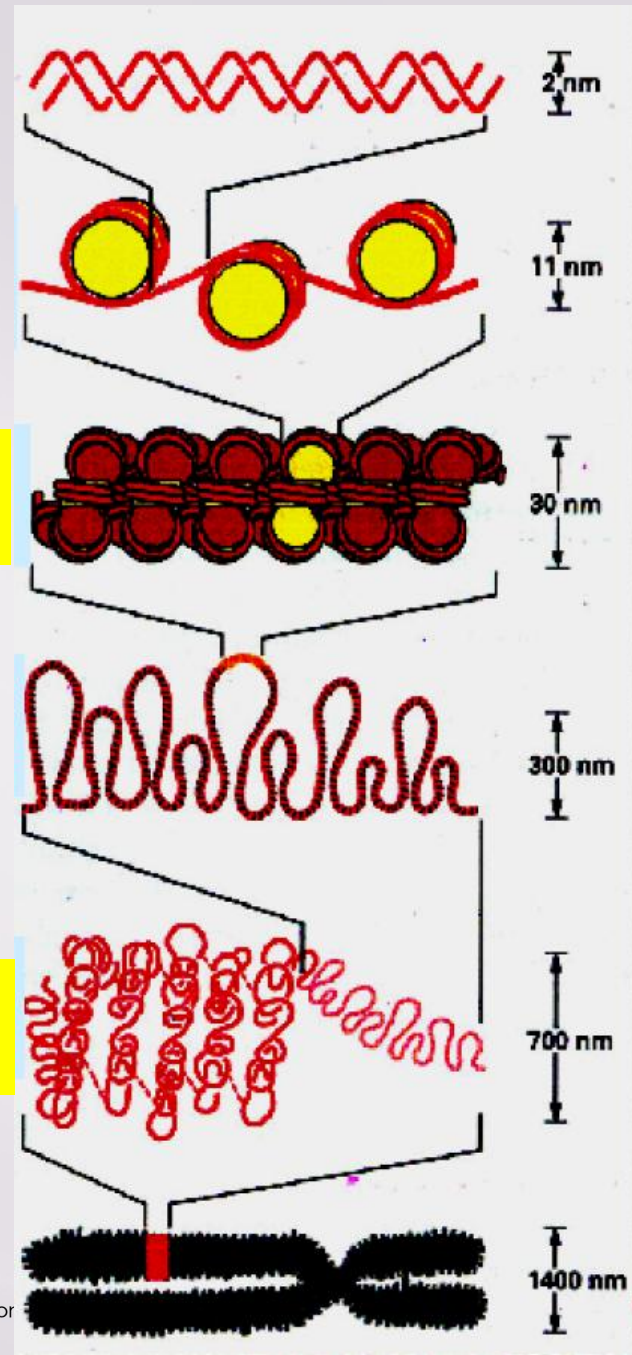
“Beads on a string” form of Chormatin

30-nm chromatin fiber of packed nucleosomes

Section of chromosome in an extended form

Condensed section of Metaphase chromosome

Entire metaphase chromosome



3.17 The Path of Nucleosomes to Chromatin Fiber and Chromosome

- The primary structure of chromatin is a **10-nm fiber** which consists of a string of nucleosomes.
- The secondary structure of chromatin is formed by interactions between neighboring nucleosomes that promote formation of more condensed fibers.
- **30-nm fibers are a prevalent type of secondary structure** that contain 6 nucleosomes/turn, organized into either a one-start solenoid or a two-start helix.
- Histone H1, histone tails, and increased ionic strength all promote the formation of secondary structures, including the 30-nm fiber.

3.17 The Path of Nucleosomes to Chromatin Fiber and Chromosome

- Secondary chromatin fibers are folded into higher-order, three-dimensional structures that comprise interphase or mitotic chromosomes.

3.18 Replication of Chromatin Requires Assembly of Nucleosomes

- Histone octamers are not conserved during replication, but H2A-H2B dimers and H3₂-H4₂ tetramers are.
- There are different pathways for the assembly of nucleosomes during replication and also independent of replication.

3.18 Replication of Chromatin Requires Assembly of Nucleosomes

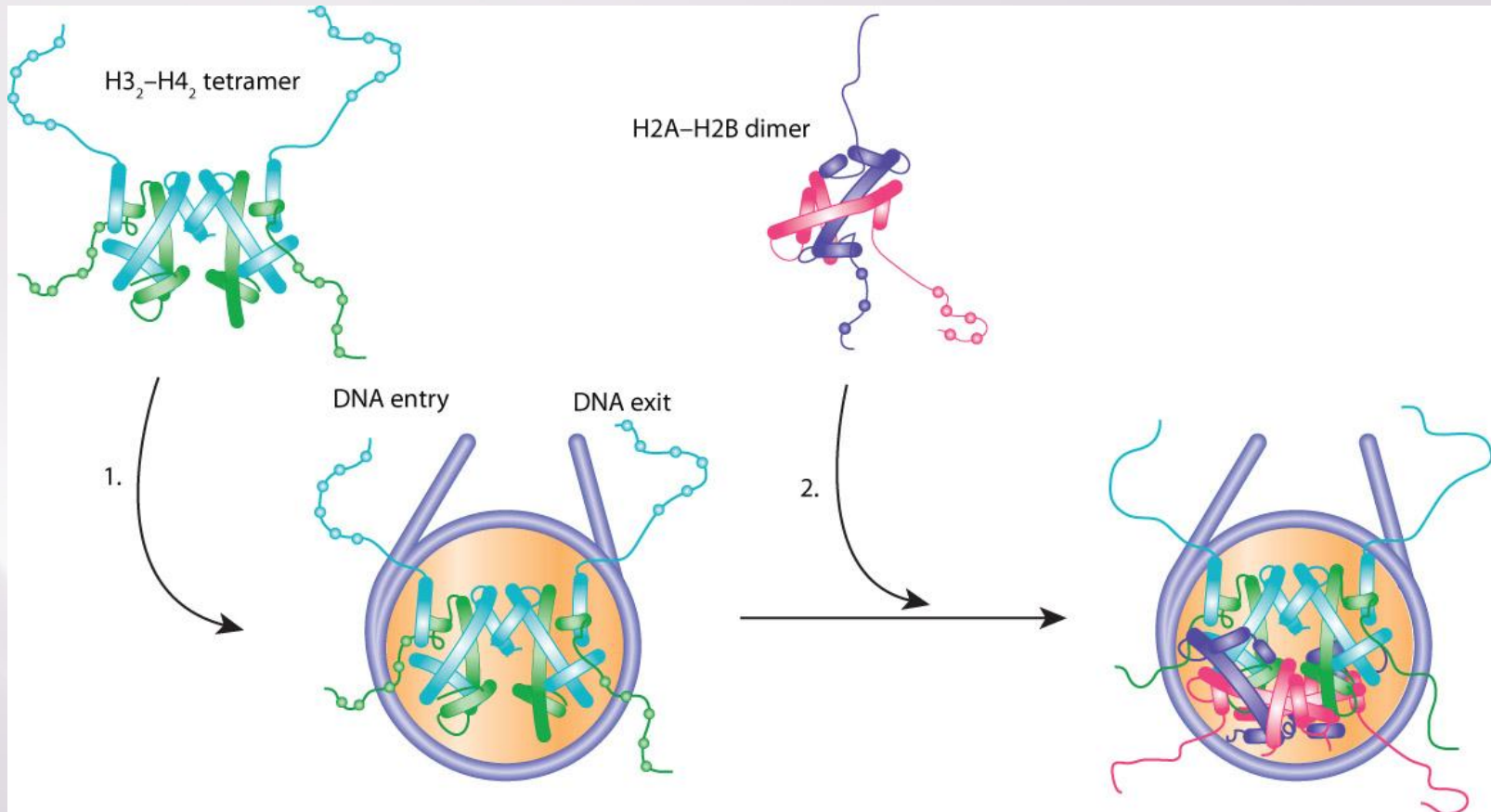


Figure : During nucleosome assembly *in vivo*, H3-H4 tetramers form and bind DNA first, then two H2A-H2B dimers are added to form the complete nucleosome.

3.18 Replication of Chromatin Requires Assembly of Nucleosomes

- Accessory proteins are required to assist the assembly of nucleosomes.
- CAF-1 and ASF1 are histone assembly proteins that are linked to the replication machinery.
- A different assembly protein, HIRA, and the histone H3.3 variant are used for **replication-independent assembly**.

3.18 Replication of Chromatin Requires Assembly of Nucleosomes

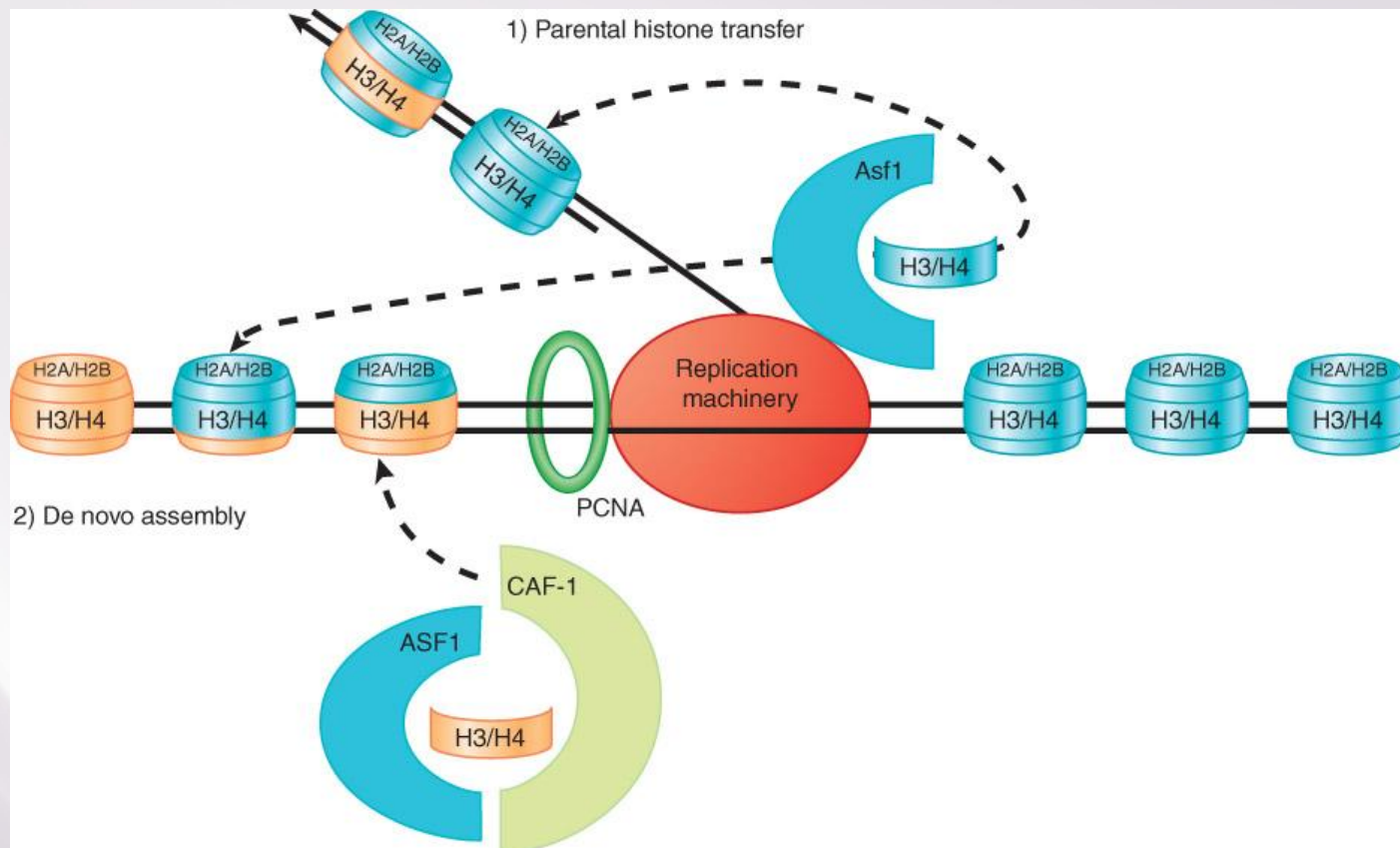


Figure : Replication fork passage displaces histone octamers from DNA.

3.19 Nucleosomes Are Displaced and Reassembled During Transcription

- Most transcribed genes retain a nucleosomal structure, though the organization of the chromatin changes during transcription.
- Some heavily transcribed genes appear to be exceptional cases that are devoid of nucleosomes.
- RNA polymerase displaces histone octamers during transcription *in vitro*, but octamers reassociate with DNA as soon as the polymerase has passed.

3.19 Nucleosomes Are Displaced and Reassembled During Transcription

- Nucleosomes are reorganized when transcription passes through a gene.
- Additional factors are required both for RNA polymerase to displace octamers during transcription and for the histones to reassemble into nucleosomes after transcription.

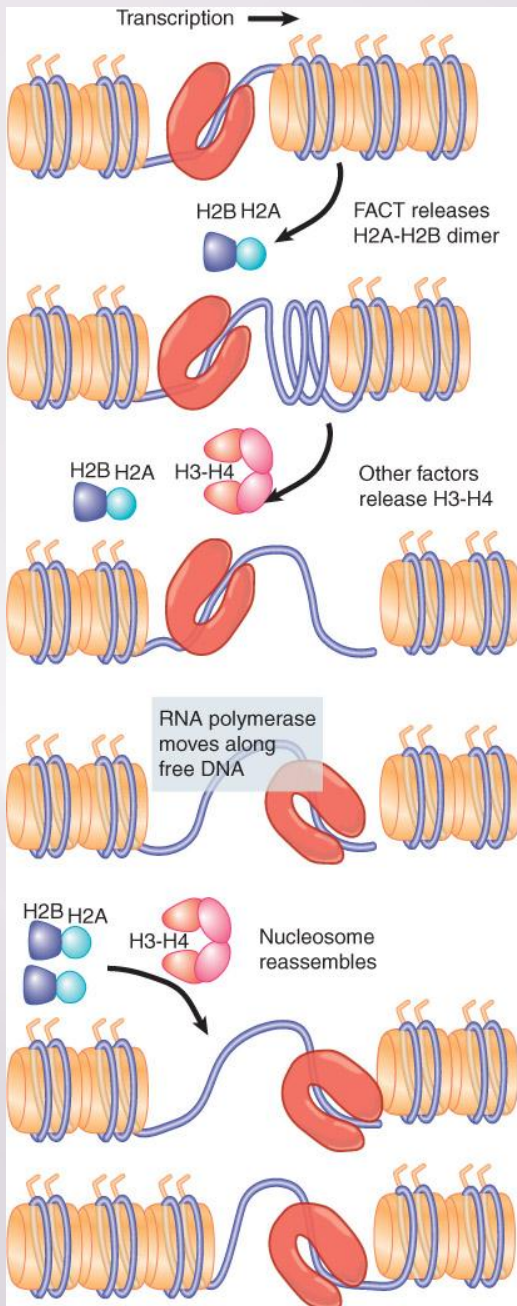


Figure : Histone octamers are disassembled ahead of transcription to remove nucleosomes.

3.20 Do Nucleosomes Lie at Specific Positions?

- Nucleosomes can form at specific positions as the result of either the local structure of DNA or proteins that interact with specific sequences.